1 (a) When the tension in the rope is more than the weight of the person, the resultant force acting on the person is opposite in direction to his velocity and the person is decelerating. B1

When the tension in the rope is less than the weight of the person, the resultant force acting on the person is in the same direction as his velocity and the person is accelerating. B1

Thus, the person will have maximum speed when the resultant force acting on him is zero and with a corresponding acceleration of zero.

(b) Man is moving at maximum speed when resultant force acting on him is zero.

$$mg = ke$$
$$e = \frac{mg}{k} = \frac{70.0 \times 9.81}{120} = 5.72 \text{ m}$$ C1

Applying Principle of Conservation of Energy,
Total energy before stepping off = Total energy at maximum speed

$$0 + 0 + 0 = \frac{1}{2}mv^2 + \frac{1}{2}ke^2 - mg(l + e)$$

$$v = \sqrt{\frac{2mg(l+e) - ke^2}{m}}$$ C1

$$= \sqrt{\frac{2 \times 70.0 \times 9.81 \times (25.0 + 5.72) - 120 \times 5.72^2}{70.0}}$$

$$= 23.4 \text{ m s}^{-1}$$ A1

(c) Applying Principle of Conservation of Energy,
GPE at highest point = EPE at lowest point

$$mg(l + e) = \frac{1}{2}ke^2$$ M1

$$70 \times 9.81 \times (25 + e) = 0.5(120)(e^2)$$

$$e = 23.6 \text{ m}$$ A1